

TidyR_homework_mcewen

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TidyR Homework

In this assignment, you will be asked to select a dataset of your choice, identify any ‘untidy’ characteristics, make it tidy, and do some exploratory analysis.

Question 1 (1 point)

Select a dataset from <https://github.com/rfordatascience/tidytuesday/tree/main/data> and import it into RStudio.

If your dataset references a paper, like the one we did in class, you can download it from its data repository. If it is only found on the tidy tuesday github, you may need to use one of these functions: `tidytuesdayR::tt_load`, `readr::read_csv`

Show me the first *10 rows* of your data (1 point).

```
package_list <- c("dplyr", "readr", "tidyverse", "tidyr", "lubridate", "dedupewider")

for (pkg in package_list) { #Installs packages in package_list
  if (!requireNamespace(pkg, quietly = TRUE)) {
    install.packages(pkg, dependencies = TRUE)
  }
  library(pkg, character.only = TRUE)
}
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats   1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
```

```
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()   masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

dls_protein_corona_size = read_csv("20250502.csv") #Reads in a raw csv file

## Rows: 44 Columns: 217
## -- Column specification -----
## Delimiter: ","
## chr (217): Type, Sample Name, S/W Version, Serial Number, Measurement Date a...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

head(dls_protein_corona_size, 10) #Outputs the first 10 rows of data

## # A tibble: 10 x 217
##   Type 'Sample Name' 'S/W Version' 'Serial Number' Measurement Date and-1
##   <chr> <chr> <chr> <chr> <chr>
## 1 Size SeNP_only_15min 1 8.02 MAL1086566 Friday, May 2, 2025 1~
## 2 Size SeNP_only_15min 2 8.02 MAL1086566 Friday, May 2, 2025 1~
## 3 Type Sample Name S/W Version Serial Number Measurement Date and ~
## 4 Size extract_only_15mi~ 8.02 MAL1086566 Friday, May 2, 2025 1~
## 5 Size extract_only_15mi~ 8.02 MAL1086566 Friday, May 2, 2025 1~
## 6 Type Sample Name S/W Version Serial Number Measurement Date and ~
## 7 Size Se_0.5mg/mL_15min~ 8.02 MAL1086566 Friday, May 2, 2025 1~
## 8 Size Se_0.5mg/mL_15min~ 8.02 MAL1086566 Friday, May 2, 2025 1~
## 9 Type Sample Name S/W Version Serial Number Measurement Date and ~
## 10 Size Se_0.2mg/mL_15min~ 8.02 MAL1086566 Friday, May 2, 2025 1~
## # i abbreviated name: 1: 'Measurement Date and Time'
## # i 212 more variables: 'Z-Average (d.nm)' <chr>, PdI <chr>,
## # 'Sizes[1] (d.nm)' <chr>, 'Sizes[2] (d.nm)' <chr>, 'Sizes[3] (d.nm)' <chr>,
## # 'Sizes[4] (d.nm)' <chr>, 'Sizes[5] (d.nm)' <chr>, 'Sizes[6] (d.nm)' <chr>,
## # 'Sizes[7] (d.nm)' <chr>, 'Sizes[8] (d.nm)' <chr>, 'Sizes[9] (d.nm)' <chr>,
## # 'Sizes[10] (d.nm)' <chr>, 'Sizes[11] (d.nm)' <chr>,
## # 'Sizes[12] (d.nm)' <chr>, 'Sizes[13] (d.nm)' <chr>, ...
```

Question 2 (3 points)

Identify at least 3 non-tidy characteristics of your data or things you would like to change in this dataset (1 point ea):

1. Each set of sample measurement duplicates is preceded by a header row, so that the column headers appear multiple times in each column.
2. The size values (in nm) and the intensity values corresponding to those sizes are contained in the same row. They should each be contained in different columns.
3. The NP concentration and replicate number is all contained in the sample name.
4. The "S/W Version" and "Serial Number" are useless information because they are the same for all samples.
5. The measurement date and time contains characters, but I would like to convert it to numeric.

Show any code you used to identify these characteristics:

Question 3 (5 points)

Fix each of your issues. Use comments to show where each of your issues listed above was fixed. Use pipes to connect fixing all of your issues together!

```
dls_protein_corona_size_cleaned = dls_protein_corona_size %>%
  filter(Type == "Size",) %>% #Removes interdigitated header rows, addresses issue 1 above
  select(-c(`S/W Version`, `Serial Number`)) %>% #Removes "S/W Version" and "Serial Number" columns, ad
  mutate(
    datetime = as.POSIXct(`Measurement Date and Time`,
                          format = "%A, %B %d, %Y %I:%M:%S %p"),
    measurement_time_numeric = hour(datetime) + minute(datetime)/60 + second(datetime)/3600) %>% #Conve
  select(-c(`Measurement Date and Time`)) %>% #Removes "Measurement Date and Time" column
  separate(`Sample Name`, into = c("sample_info", "replicate"), sep = " ") %>%
  mutate(
    replicate = as.integer(replicate), #Extracts replicate information and adds it to the new column "r
    concentration = str_extract(`sample_info`, "(?<=_)\\d+\\.?.?\\d*(?=mg/mL)") #Extracts NP concentration
  )

sizes_and_intensities = dls_protein_corona_size_cleaned %>%
  select(c(`sample_info`, starts_with("Sizes"), starts_with("Intensities")))) #Creates a new dataframe s

colnames(sizes_and_intensities)[72:141] = sizes_and_intensities[1, 2:71] #Renames column headers of int

sizes_and_intensities_cleaned = sizes_and_intensities %>%
  select(-c(starts_with("Sizes"))) %>% #Removes repetitive size data
  pivot_longer(cols = -1, names_to = "size_nm", values_to = "intensity_percent") #Pivots table so that

sizes_and_intensities_cleaned$size_nm = as.numeric(sizes_and_intensities_cleaned$size_nm) #Converts "si
sizes_and_intensities_cleaned$intensity_percent = as.numeric(sizes_and_intensities_cleaned$intensity_pe

dls_protein_corona_size_concentration = dls_protein_corona_size_cleaned %>%
  select(-c(starts_with("Intensities"), starts_with("Sizes"), starts_with("Volumes"))) %>% #Creates a n
  drop_na(`concentration`) %>% #Drops all rows with NA in the "concentration" column, as these are cont
  mutate(elapsed_time = `measurement_time_numeric` - `measurement_time_numeric`[1]) #Computes the elaps

dls_protein_corona_size_concentration$`Z-Average (d.nm)` = as.numeric(dls_protein_corona_size_concentra
dls_protein_corona_size_concentration$PdI = as.numeric(dls_protein_corona_size_concentration$PdI) #Conv
dls_protein_corona_size_concentration$concentration = as.factor(dls_protein_corona_size_concentration$c
```

Question 4 (2 points)

List a question you would like to answer using this dataset:

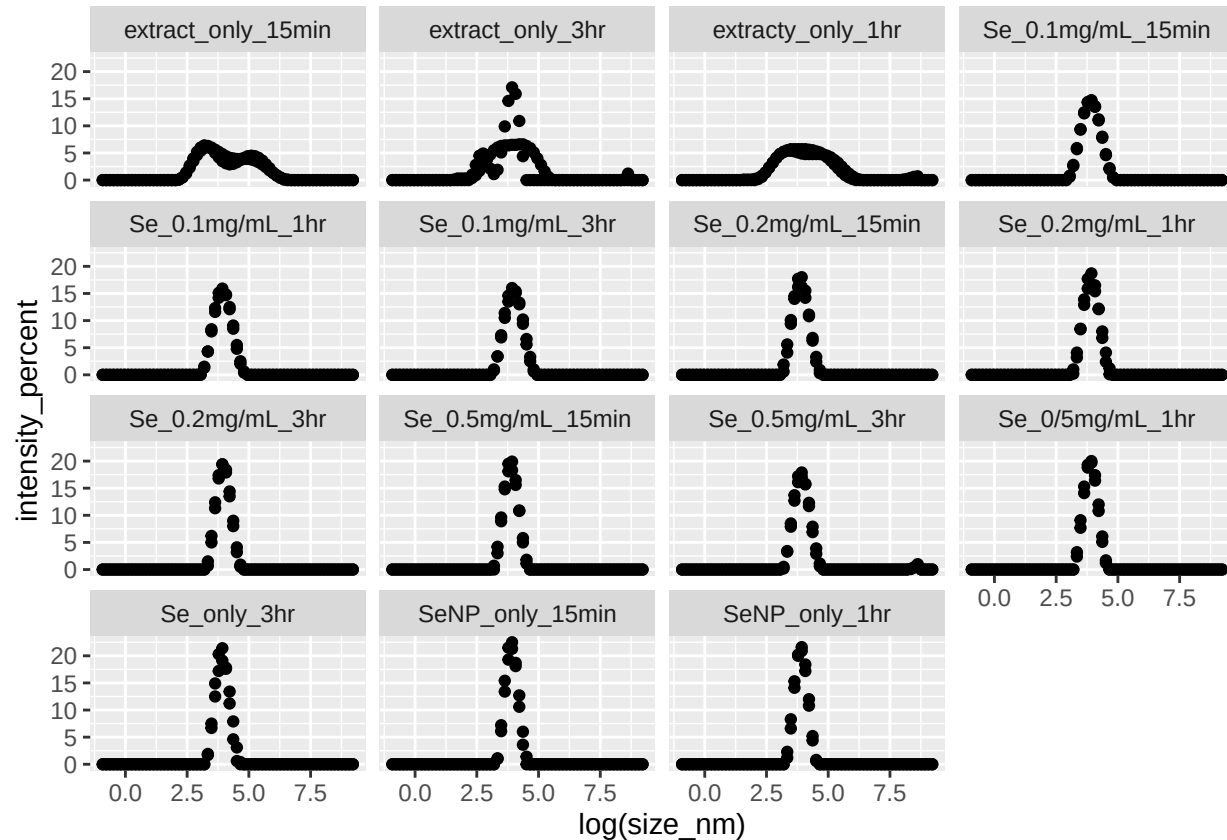
How does the average hydrodynamic particle diameter change as a function of incubation time and as a function of protein concentration?

Question 5 (5 points)

Using stats, plots, or both, answer your question. Make sure you output any stats or plots you create in this code chunk (i.e., that it shows up in the knitted document).

NOTE: you do not have to use a formal stats test if you are not familiar! Feel free to just use summary stats and/or figures to answer your question.

```
sizes_and_intensities_cleaned %>%
  ggplot(aes(x=log(size_nm), y=intensity_percent)) +
  geom_point() +
  facet_wrap(~ sample_info)
```

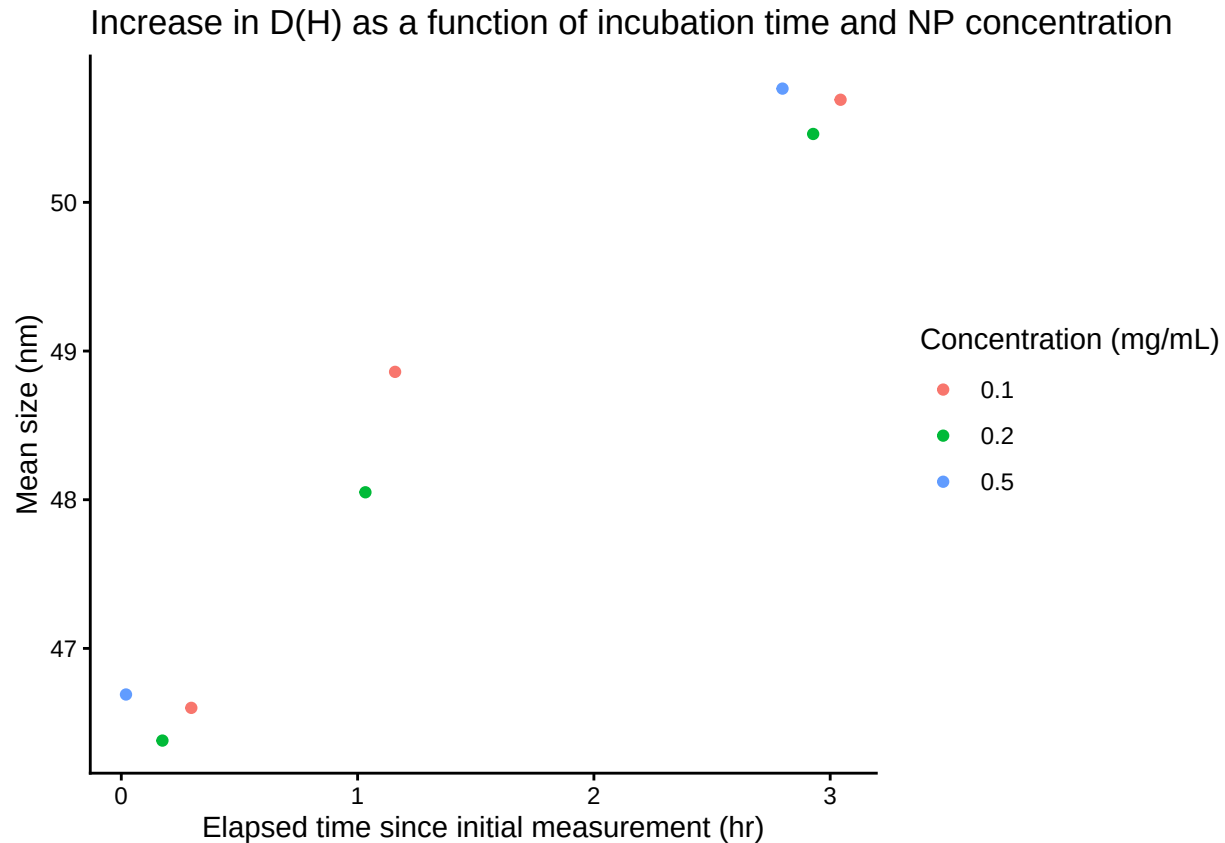


```
dls_protein_corona_size_concentration %>%
  group_by(sample_info, concentration) %>%
  summarise(mean_size = mean(`Z-Average (d.nm)`),
            mean_time = mean(elapsed_time),
            sd_size = sd(`Z-Average (d.nm)`),
            sd_time = sd(elapsed_time),
            n = n(),
            se_size = sd_size/sqrt(n),
            se_time = sd_time/sqrt(n)) %>%
  ggplot(aes(x=mean_time, y=mean_size, color = concentration)) +
  geom_point() +
  labs(x = "Elapsed time since initial measurement (hr)",
       y = "Mean size (nm)",
       title = "Increase in D(H) as a function of incubation time and NP concentration",
       color = "Concentration (mg/mL)") +
  theme(plot.title = element_text(hjust = 0.5), plot.subtitle = element_text(hjust = 0.5)) +
  theme_classic()
```

'summarise()' has regrouped the output.

i Summaries were computed grouped by sample_info and concentration.

```
## i Output is grouped by sample_info.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(sample_info, concentration))' for per-operation
##   grouping ('?dplyr::dplyr_by') instead.
```



Provide the answer (in words) to your question below:

For each concentration tested, the mean particle size (as measured by hydrodynamic radius) increases as incubation time increases. It appears to plateau, but longer incubation times are required to confirm this.

Question 6 (4 points)

Find two functions we haven't introduced you to that are a part of tidyR, dplyr, or other related tidyverse packages. Provide the function name (including the package it's from) and what it does.

- 1. The `relocate()` function from the `dplyr` package will change the position of columns without removing them.
- 2. The `replace_na()` function from the `tidyR` package will replace all NA values with a specific, known value